

Q1. > WAP to check if a number is even or odd.

```
import java.util.Scanner;
```

```
public class Evenodd {  
    public static void main(String args[]) {  
        Scanner sc = new Scanner(System.in);  
        SOP("Entre a number");  
        int num = sc.nextInt();  
        if (num % 2 == 0) {  
            SOP("It is even");  
        } else {  
            SOP("It is odd");  
        }  
    }  
}
```

Output

Entre a number

8

It is even

Q2→ To check if a given number is Palindrome (Python):

```
a = input("Entre a number")  
b = [::-1]
```

```
if (a == b):  
    print("It is Palindrome")  
else:  
    print("Not Palindrome")
```

Output:

Entre a number

121

Palindrome

Q3→ To calculate factorial?

```
a = int(input("Entre a number"))
```

```
i = 1, m = 1
```

```
while (i <= a):
```

```
    m = m * i
```

```
    i = i + 1
```

```
print(m)
```

Output

Entre a number

3

→

6

QD Calculating Student's result

```
class Student:
```

```
    def __init__(self, name, marks):  
        self.name = name
```

```
        if len(marks) != 3:
```

```
            print("less subjects")
```

```
            self.marks = marks
```

```
            self.total = 0
```

```
            self.average = 0
```

```
    def Cal(self):
```

```
        self.total = sum(self.marks)
```

```
        self.average = (self.total) /  
                        (len(self.marks))
```

```
    def dis(self):
```

```
        print(self.name)
```

```
        print(self.marks)
```

```
        print(self.total)
```

```
        print(self.average)
```

~~Student1 = Student(name, marks)~~

name = "Aditya"

marks = [99, 99, 99]

S1 = Student(name, marks)

~~S1.calculate_result()~~

S1.calC()

S1.dis()

Output:

Aditya

[99, 99, 99]

~~99~~

~~297~~

99

Q5. > Add 2 integers:

```
import java.util.Scanner;  
class Add {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    Add() {
```

```
        S.O.P("Entre a num:");
```

```
        int a = sc.nextInt();
```

```
        S.O.P("Entre a num:");
```

```
        int b = sc.nextInt();
```

```
        S.O.P("Sum: " + (a+b));
```

```
    }
```

```
    public class Main {
```

```
        public static void main(String  
            args)
```

```
        {  
            Add a = new Add();
```

```
        }
```

Output:

Entre a num: 9

Entre a num: 9

Sum: 18

Q6.)

Swapping 2 numbers:

```
class Swapper {  
    void swap(int a, int b) {  
        SOP("Before a=" + a + ", b=" + b);  
        int temp = a;  
        a = b;  
        b = temp;  
        SOP("After a=" + a + ", b=" + b);  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Swapper s = new Swapper();  
        s.swap(10, 20);  
    }  
}
```

Output:

~~Before a=10, b=20~~~~After a=20, b=10~~

Q7.) To check if a number is even or odd:

```
class ioe {
```

```
    void ioe (int a) {
```

```
        if (a == 0) {
```

```
            SOPC("Even");
```

```
        }
```

```
        elseif (a % 2 == 0) {
```

```
            SOPC("Even");
```

```
        }
```

```
    } else {
```

```
        SOPC("Odd");
```

```
    }
```

```
}
```

```
}
```

```
class OddEven {
```

```
    public static void main (String[] args) {
```

```
        ioe a = new ioe();
```

```
        a.ioe(6);
```

```
    }
```

```
}
```

Output
Even

Q80) class Large {
void large(int a, int b, int c){
if (a > b && a > c){
SOP("a is largest");
}
else if (b > a && b > c){
SOP("b is largest");
}
else {
SOP("c is largest");
}
}
}

public class Largest {
public static void main(String args){
Large obj = new Large();
obj.large(3, 4, 5);
}
}

Output:

C is ~~the~~ largest

Q90)

Calculator Using Switch:

import java.util.Scanner;

public class calS

{
 public static void main (String[] args){
 Scanner sc = new Scanner(System.in);

SOP("Enter first num");

double a = sc.nextDouble();

SOP("Enter Second num");

double b = sc.nextDouble();

SOP("1 For Add");

SOP("2 For Subtract");

SOP("3 For multiply");

SOP("4 For divide");

~~SOP("5 For Exit")~~

int c = sc.nextInt();

switch(c){

case 1:

SOP(a+b);

break;

case 2:

SOP(a-b);

break;

case 3:

SOP(a*b);

break;

case 4:

```

SOP(a/b);
break;
if (b != 0) {
    SOP(a/b);
} else {
    SOP("error");
}
break;

```

de Fault :

```

    SOP("Invalid option");
}

```

}

Output

Entre first num

4

Entre Second num

5

1 for Add

2 for Subtract

3 for Multiply

4 for divide

1

Mg
24/9/25

Q10.) WAP to Sum of values of an array.

```
import java.util.Scanner;  
class ARR2
```

```
    void arr() {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the length");
```

```
        int m = sc.nextInt();
```

```
        int[] arr = new int[m];
```

```
        int i = 0;
```

```
        while (i < m) {
```

```
            SOP("Enter the element");
```

```
            int b = sc.nextInt();
```

```
            arr[i] = b;
```

```
            i = i + 1;
```

```
        }
```

```
        i = 0;
```

```
        int sum = 0;
```

```
        SOP("Sum is :");
```

```
        while (i < m) {
```

```
            sum = sum + arr[i];
```

```
            i = i + 1;
```

```
        }
```

```
        SOP(sum);
```

```
    }
```

```
public class progas  
    public static void main (String[] args)  
    {  
        ARR d = new ARR(3);  
        d.arr();  
    }  
}
```

Output :

Enter the length

3

Enter the element

1.

Enter the element

2

Enter the element

3

Sum is :

6

Q11) import java.util.Scanner;
class ARR {
 void arr() {
 Scanner sc = new Scanner
 (System.in);
 SOP("enter the length");
 int n = sc.nextInt();
 int[] arr = new int[n];
 int i = 0;
 while (i < n) {
 SOP("Enter the element");
 int b = sc.nextInt();
 arr[i] = b;
 i = i + 1;
 }
 int sum = 0;
 while (i < n) {
 ~~sum~~ sum = sum + arr[i];
 i = i + 1;
 }
 SOP("average is");
 SOP(sum/n);
 }
}

Output:

Enter the length:

3

Enter the element:

1

Enter the element:

2

Enter the element:

3

Average is:

2

Q12) WAP to create a class Student with USN, name an array of marks etc.

```
import java.util.Scanner;  
class Student {
```

```
    int sum;
```

```
    int ns;
```

```
    Scanner sc = new Scanner(System.in);
```

```
    int details () {
```

```
        SOP("enter your name");
```

```
        String[] name = sc.nextLine().split(" ");
```

```
        SOP("enter your USN");
```

```
        String[] usn = sc.nextLine().split(" ");
```

```
        SOP("enter the no of Subjects");
```

```
        int ns = sc.nextInt();
```

```
        sc.nextLine();
```

```
        String[] subjects = new String[ns];
```

```
        int[] marks = new int[ns];
```

```
        int i = 0;
```

```
        int sum = 0;
```

```
        while (i < ns) {
```

```
            SOP("enter the name of Subject");
```

```
            subjects[i] = sc.nextLine();
```

```
            sc.nextLine();
```

~~String~~ []
~~int~~ marks

```
SOPC "Entre marks for" + Subject[i]
marks [i] = sc.nextInt();
sum = sum + marks[i];
i = i + 1;
```

```
SOPC "Sum is : " + sum);
return sum;
```

§ }
~~void~~

```
void cal (int sum)
```

```
{
    int credit = 4;
```

```
    int sgpa = sum / 4;
```

```
    SOP ("The sgpa is : " + sgpa);
}
```

g

}

```
Public class main prog2
```

```
{
    Public static void main
```

```
(String[] args)
```

```
{
    int sum;
```

```
    Student stud1 = new Student()
```

```
    sum = stud1.details();
```

```
    stud1.cal(sum);
}
```

}

}

Output:

enter your name

Aditya

enter your USN

1CN24CS021

enter no of Subjects

3

enter the name of Subject

Maths

Marks for Maths : 90

enter the name of Subject

Python

Marks for Python : 90

enter the name of Subject

C

Marks for C : 90

~~Sgpa is : 90~~

MG
8/10/21

Q13) Bank operation class

```
import java.util.Scanner;
```

```
class Account {
```

```
    protected String customerName;  
    protected String accountNumber;  
    protected String accountType;  
    protected double balance;
```

```
    public Account(String customerName,  
                    String accountNumber, String  
                    accountType, double initialBalance)  
    {
```

```
        this.customerName = customerName;  
        this.accountNumber = accountNumber;  
        this.accountType = accountType;  
        this.balance = initialBalance;
```

```
    }
```

```
    public void deposit(double amount)
```

```
    {
```

```
        if (amount > 0) {
```

```
            balance += amount;
```

```
            SOP (amount + " deposited.
```

```
            New balance: " + balance);
```

```
        }
```

HP
15/12/25

else {

SOPC("Invalid deposit amount")

}

}

public void displayBalance() {

SOPC("Account Balance: " + balance)

}

public void displayBalance() {

SOPC("Account Balance: " + balance)

}

public boolean withdraw(
double amount) {

if (amount > 0 && amount
≤ balance) {

balance -= amount;

SOPC(amount + " withdrawn")

New balance

}

else {

SOPC("less balance");

return false;

}

}

}

```
class SavAcct extends Account {
    private double interestRate;
```

```
    public SavAcct(String customerName, String accountNumber, double initialBalance, double interestRate) {
```

```
        super(customerName, accountNumber, "Savings", initialBalance);
        this.interestRate = interestRate;
```

```
    }
```

```
}
```

```
class CurrAcct extends Account {
    private double minimumBalance;
    private double servicecharge;
```

```
}
```

```
    public void issueChequeBook() {
        SOP("It is issued" + customerName);
```

```
}
```

```
class PublicBank
```

```
    public static void main(
        String [] args) {
```

account = new SaveAcct()

Output :

Name: Adi

Acc Number: 123

1> Deposit

2> Withdraw

1000

1000

1000 Deposited

3Q

create a class Book which contains 4 members: name, author, price, num-pages. Include a constructor to set the values for the members. Include methods to set and get the details of the objects. Include a toString() method that could display the complete details of the book. develop a java program to create n book-objects.

```
import java.util.Scanner;
```

```
class Book {
```

```
    public String name;
```

```
    public String author;
```

```
    public double price;
```

```
    public int numpages;
```

```
    public Book (String name,  
                String author, double price,  
                int numpages) {
```

```
        this.name = name;
```

```
        this.author = author;
```

```
        this.price = price;
```

3. This num pages = num pages;

```
public void setAuthor (String  
name) {  
    this.name = name;  
    author = author;
```

```
public void setName(String  
    name) {  
    this.name = name;  
}
```

```
public void setPrice (double price)
    this price = price;
```

```
public void setNumPages (int numPages) {  
    this.numPages = numPages;  
}
```

```
public String getAuthor() {  
    return author;  
}
```

```
public String getName() {  
    return name;  
}
```

```
public double getPrice() {  
    return price;  
}
```

```
public int getNumPages() {  
    return numPages;  
}
```

```
public String toString() {  
    return "Book Name: " + name +  
        "\nAuthor: " + Author +  
        "\nPrice: " + Price +  
        "\nNumber of Pages: " +  
        numPages;  
}
```

```
public class BookDemo {  
    public static void main  
        (String[] args) {  
        Scanner sc = new Scanner  
            (System.in)
```

Page _____

```
SOP("Enter no of book:");  
int m = scanf("%d", &m);
```

```
scanf("%d", &m);
```

```
Book[] books = new Book[m];
```

```
for (int i = 0; i < m; i++) {
```

```
    SOP("Enter details  
        of Book " +  
        (i + 1));
```

```
    SOP("Name:");
```

```
    String Name = sc.nextLine();
```

```
    SOP("Author:");
```

```
    String author = sc.nextLine();
```

```
    SOP("Price:");
```

```
    double price = sc.  
        nextDouble();
```

```
    SOP("Num of pages:");
```

```
    int pages = sc.nextInt();
```

```

books[i] = new Book(name
                    , author, price
                    , pages);
    }

```

```

    SOP("\n --- Books ---");
    for (i = 0; i < n; i++) {
        SOP("\n Books + (i+1)");
        SOP(books[i].toString());
    }
}

```

Output:

Enter the number of Book: 1

Name: cool

Author: adi

Price: 400000

Number of Pages: 1

-- Book Details --

Book 1 Details

Name: cool

Author: adi

Price: 400000

Number of Pages: 1

Handwritten signature/initials

Q6 create a package CIE which has two classes - Personal and Internal. The class Personal has members like usn, name, sem. The class Internal has an array that stores the internal marks scored in five courses of the current semester of the student. Create another package see which has the class External which is a derived class of Personal. This class has an array that stores the see marks scored in five courses of the current semester of the student. Import the 2 packages in a file that declares the final marks of n students in all five courses.

Final Marks

```

-> CIE \
    |
    | -> Personal
    | -> Internal
    |
-> SEE \
    |
    | -> External
  
```

CIE \

```

import java.util.Scanner;
public class Personal {

```

~~Personal~~

```

    int usn;
    String name;
    int sem;

```

```

    Personal(int usn, String
               name, int sem)
    {

```

```

        this.usn = usn;

```

```

        this.name = name;

```

```

        this.sem = sem;
    }
}

```

Public class Internals & extends
Personal &

Scanner sc = new Scanner(
System.in);

int [100][5] marks;

Internals (~~enter your~~
int usm)

{
for (int i = 0; i < 5; i++)

{
marks[usm-1][i]
= sc.nextInt();

}

}

void display() {
int usm

i = 0;

for (i = 0; i < 5; i++)

{
SOP("Enter marks");

SOP(marks[usm-1][i]);

See\

CIE, Personal

Import ~~See~~ ~~External~~

Import java.util.Scanner;

Public class External extends
Personal {

Scanner sc = new Scanner(System.in)

int [100][5] seemarks;

getExternal (int usn)

{

for (int i=0; i<5;

i++) {

usn-

seemarks[~~usn~~][i]

= sc.nextInt();

}

}

void displaySee (int usn) {

for (int i=0; i<5; i++)

{ S.O.P ("See Marks In");

S.O.P (seemarks[usn-1]

[i]);

}

Final Marks

Import CIE. * ;

Import SEE. * ;

Public class ~~Main~~ FinalMarks

Public static void main
(String [] args) {

~~Personal~~ Personal s1 = new
Personal(1, "Adi",
3);

Internals (1);

~~External~~ External (1);

s1.displayCIE();

s1.displaySEE();

Output:

95

95

95

95

85

60

60

60

90

100

CIE Marks:

95

95

95

95

85

See Marks:

60

60

60

90

100

12/11/25

Q A ride-sharing app has different ride types, Mini, Prime, SUV, each type has different fare per km.

Interface Ride

double calculatefare
(double distance)

String getRideType();

}

import java.util.Scanner;
class Mini extends Ride

double
calculatefare
(double dis)

Mini()

int fare = 1;

Scanner sc = new Scanner
~~calculate~~ (System.in);

return (fare * dis);

}

String getRideType()

scop (getRideType());

scop ("type distance")

return ("You have
chosen
Mini");

double a = sc.next
Double();

}

```
SOP(Calculatefare(a));  
}  
}
```

```
class Prime extends Ride {
```

```
Prime() {  
    in fare = 5;
```

```
double Calculatefare(double dis)  
{  
    return(fare * dis);  
}
```

```
String getRideType() {
```

```
    return("You have Chosen  
    Prime");  
}
```

```
SOP(getRideType());  
double a = scanner.nextDouble();  
SOP(Calculatefare(a));
```

```
}
```

```
}
```

```
class SUV extends Ride {  
    int fare = 10;  
    double calculatefare(double dis)  
    {  
        return (fare * dis);  
    }  
}
```

```
String getRideType() {
```

```
    return ("You have chosen  
        SUV");  
}
```

```
SOP(getRideType());
```

```
double a = sc.nextdouble();
```

```
SOP(calculatefare(a));
```

```
}
```

```
class Main {
```

```
    public static
```

```
void SOP() { choose 1 for Mini, 2 for Prime  
    int i = sc.nextInt();  
}
```

```
switch(i) {
```

```
    case 1:
```

```
        Mini obj1 = new Mini();  
        obj1.Mini();  
        break;
```

```
    case 2:
```

```
        Prime obj1 = new Prime();  
        obj1.Prime();  
        break;
```

```
    case 3:
```

```
        SUV obj1 = new SUV();  
        obj1.SUV();  
        break;
```

```
    case 0:
```

```
        exit(1) exit(1)
```

```
} while (i != 0);
```

```
}
```

Output:

Q> Exception program :-

~~class Exception {~~

class WrongAge extends
Exception {

WrongAge(String msg
{

super(msg);

}

}

~~public class Main {~~

~~public static void main
(String[] args)~~

~~{~~

~~try {
 SOP("Enter
 int age");~~

```
class dad {  
    int dAge;  
    dad() {  
        SOP("Entre dad's age");  
        dAge = SC.nextInt();  
        try {  
            if (dAge < 0) {
```

```
                throw new  
                WrongAge("negative age");  
            }  
        }  
    }  
}
```

```
    } catch (WrongAge e) {  
        SOP(e.getMessage());  
    }  
}
```

```
class son {  
    int sAge;  
    son() {
```

```
        SOP("Entre son's age");  
        sAge = SC.nextInt();  
        try {  
            if (sAge < 0) {
```

```
                throw new  
                WrongAge("wrong age");  
            }  
        }  
    }  
}
```

```
} SOP(e.getMessage());
```

```
Public class main {
```

```
    PSVM (String [] args) {
```

```
        try {
```

```
            Dad d = new Dad();  
            d.Dad();
```

```
            Son s = new Son();  
            s.Son();
```

```
            if (d.dage < s.dage)
```

```
            {
```

```
                throw new  
                WrongAge ("Wrong age")
```

```
            } catch (WrongAge e) {
```

```
                SOP(e.getMessage());
```

```
            }  
        }  
    }  
}
```

Output

Enter dad's age

64

Enter son's age

-1

Wrong Age

~~8~~

Prog - 8

WAP using Thread to Print Statement w
class CollegeThread extends Thread

public void run() {

try {

while (true) {

SOPC " BMS College

Thread.sleep(1000)

} catch (InterruptedException
e) {

SOPC " Interrupted"

}
}
}

class CSEThread extends Thread

public void run() {

try {

while (true) {

SOPC " CSE "

Thread.sleep(2000)

}
}
}

```

        catch (InterruptedException e) {
            SOPC("InterruptedException");
        }
    }
}

```

```

public class Main {
    public static void main(
        String[] args) {

```

```

        CollegeThread t1 = new
            CollegeThread();

```

```

College
        CSEThread t2 = new
            CSEThread();

```

```

        t1.start();
        t2.start();

```

```

    }
}

```

Output:

10/12/21 BMS college of ~~Engineers~~ in C++ (10 sec)
 CSE (in 1 sec)
 (in 2 sec) 9/12